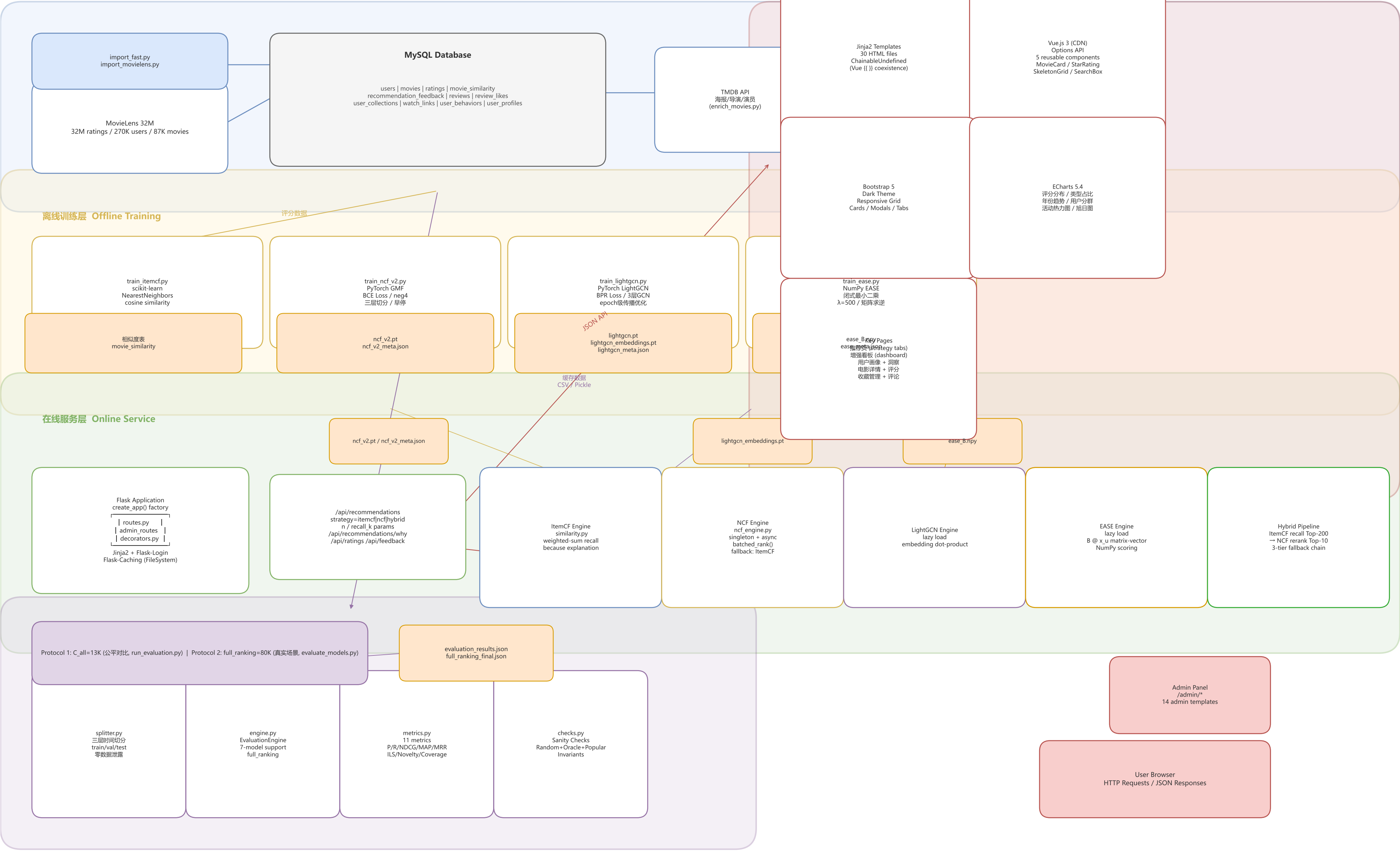


图 4-1 系统总体架构



(1) MovieLens → import → MySQL

(2) MySQL → train_*.py → artifacts (.pt/.npy/sims)

(3) artifacts → Flask API → JSON recommendations

(4) JSON → Jinja2/Vue.js → rendered pages

(5) MySQL → evaluation engine → evaluation_results.json